

Easwaran on Rational Norms

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1. A less-demanding interpretation of probabilism

On a common way of understanding “probabilism”, it’s a bunch of claims about the normative / rational significance of *objective logical relationships* among propositions. E.g.:

- If P is (‘logically’) *necessary*, your credence in it should be 1.
- If P (‘logically’) *entails* Q, your credence in P should be no more than your credence in Q.
- If P and Q are (‘logically’) *incompatible*, both entail R, and R entails everything they both entail, your credences in P and Q should sum to your credence in R.

On this interpretation, probabilism entails controversial claims of *logical omniscience*.

E.g.: for any P and Q, your credence in [long tautology involving P, Q, $\wedge$, $\neg$] should be 1. This leads to a well-known worry about over-demandingness.

Kenny proposes, instead, to understand probabilism in terms of certain *relations between the agent* and the propositions in question.

- Prime example: rather than looking at what is *in fact* possible (in some or other objective sense), we should consider what the agent “finds” possible, or what is “doxastically possible” for them. Even logical contradictions can be found possible. So long as the agent *finds* [negation of long tautology] possible, failing to have credence 1 in it need not violate Kenny-style probabilism.

The agent has a set of doxastic possibilities, each of which corresponds to a set of propositions (the propositions true at it). A doxastic possibility need not be closed under entailment (in any objective sense)—e.g. P and Q could both true at it without $P \wedge Q$ being.. In some cases, the set may be logically inconsistent.

Kenny-style probabilism only requires, e.g., that when the doxastic possibilities where R is true are exactly those where one of P and Q is true, and there is no doxastic possibility at which both P and Q are true, then one’s credence in R equals the sum of one’s credences in P and Q.

- *Note:* Kenny leaves it open that logical omniscience in the ordinary sense might also be a “rational requirement”. But he maintains that if it is, ‘the nature of that rational requirement is quite different from the rational requirement of probabilism’ (5).

2. A worry: does this interpretation of probabilism demand *anything*?

Fact: for any function f from propositions to $[0,1]$, there is a set S of sets of propositions, a σ -algebra Σ on S , and a probability function $\Pr$ on Σ such that for every proposition p , $f(p) = \Pr\{X \in S \mid p \in X\}$.

Proof: let S contain, for each $y \in [0,1]$, the set $X_y = \{p : f(p) \geq y\}$. Let Σ contain, for each $z \in [0,1]$, the set $T_z = \{X_y \mid y \leq z\}$. Let $\Pr(T_z) = \sup\{y \mid X_y \in T_z\}$. Extend this to the smallest σ -algebra containing all the T_z .

So unless we can get an *independent* grip on the relevant agent-relativized relations like ‘finding possible’—independent of what the agent’s credences actually are—this kind of probabilism threatens to be trivial (impossible to violate).

3. Dutch Book and accuracy arguments for (Kenny-style) probabilism

1. Basic principle: ‘a cognitive state is rational to the extent that it does a good job of playing its functional role’.
2. ‘the functional role for degree of belief [...] is to successfully guide actions that result in the satisfaction of the agent’s desires or promotion of its values.’
3. By the Dutch Book theorem, if an agent’s degrees of belief are not probabilistic, then ‘there are actions the agent prefers that collectively guarantee [i.e. are such that the agent finds it necessary] that things go badly according to her values’.

Conclusion: if an agent’s degrees of belief are not probabilistic, the agent is irrational.

- **Q:** 1 and 2 seem *prima facie* like they imply highly *externalistic* view of rationality for degrees of belief—a conception on which agents who in fact get what they value have rational degrees of belief, and those who don’t, don’t. But Kenny’s favourite “conception” of rationality is super-internalistic and super-permissive. Can we understand 1 and 2 in a way that’s compatible with this?
- **Q:** what is the notion of “functional role” here? Is it the notion used in *functionalism*, that of a causal role that is *constitutive* of having the mental state in question? Or is it a teleological notion as in ‘the function of the heart is to pump blood’?

Kenny also gives a Joyce-style accuracy argument for probabilism. He suggests that it too ‘shows that a violation of probabilism is irrational, since [it] guarantees bad results according to one’s own values’.

- **Q:** what about an agent who cares entirely about, say, happiness (their own, or everyone’s), and only values accuracy insofar as it’s a means to the promotion of happiness. Does the accuracy argument apply to such an agent?

4. Diachronic and interpersonal norms

Kenny thinks that there is a rational requirement that agents ‘plan to update their degrees of belief in line with conditionalization’ which is of the same sort as, and can be argued for in the same way as, the rational requirement to obey probabilism. But this is not a “diachronic norm”: it just concerns the relations between two different mental states one can have at a time, credence and planning (**Q**: is this the same as *intention*?).

A *truly* diachronic norm would be something different—some normative fact that is sensitive to the facts of personal identity over time. Kenny considers two challenges to the existence of such norms.

Parfit’s challenge: personal identity is too vague / metaphysically shallow to carry any normative weight.

Christensen’s challenge: if Dutch Book arguments for diachronic norms were any good, they would also support structurally isomorphic interpersonal norms (e.g. requiring spouses / bank-account-sharers to have the same credence function).

Kenny’s “shared interests” proposal: there *are* norms (of a different sort than those of probabilism and logical omniscience) that require certain kinds of “coherence” between the credence function of x at t and the credence function of y at t' to the extent that x at t has (a) “shared interests” with y at t' , and (b) the “capacity to co-ordinate” with y at t' . The “shared interests” and “capacity to co-ordinate” relation often obtain to a high degree when $y=x$ and t' is close to t . In that case we get (approximately?) a requirement that y ’s credence function at t' derive from x ’s credence function at t by conditionalization on some proposition. But it also obtains (to a lower degree) between distinct people, e.g. spouses or members of a single scientific community.

- **Q**: how will the shared-interest and capacity-to-coordinate relations actually show up in the theory of rationality?
 - Model A: On a common way of thinking about the normative status of conditionalization, facts about what credence function one *in fact had* in the past make a difference to what’s rational now. There could have two people who are mental duplicates at t one of whom is rational and the other of whom isn’t, because the former but not the latter has credences derived from earlier credences by conditionalization). One could, correspondingly, think that when A and B are in the same community, the credences A *in fact has* make a difference to whether B’s credences are rational, and the credences B *in fact has* make a difference to whether A’s credences are rational? But this seems kind of weird!
 - Model B: in the literature on “peer disagreement”, the usual picture is that what matters to an agent’s rationality is what an agent *thinks* about the credences of other relevantly related agents (and about how they are related to them), not

what those credences in fact are (or how they are in fact related). But this seems odd in the temporal case: why should it matter to my current rationality whether my current credences fit with what I *think* my past credences were? Also, this model doesn't really generate diachronic requirements, just more synchronic requirements relating one's credences about some subject matter to one's credences about other people's credences about that subject matter (and the extent of their shared interests).

- Model C: failures of coherence are not failures of rationality for any individual, only of the group. The group is rationally required to be such that its members' credences relate in a certain way, but that doesn't imply anything about what the members are required to do.
 - Q: what is the relation that the group is required to maintain among its members credences, if not identity? Is it OK or irrational if some people in the group conditionalize on some evidence that they don't share with the other members of the group? If the latter: does irrationality begin as soon as they get the new information, or do they only have to share it as soon as it's practical? What if the sharing would require an expensive international phone call that would frustrate the group's other preferences?
 - One worry about this is that it's unclear what it even means in the temporal case? Do we have to say things 'Fred was irrational yesterday, although he was rational at every instant yesterday?'

5. Reconstruing probabilism as an "inter-fragmental" norm

In the last section of chapter 7, Kenny entertains a vision on which the rational requirement of probabilism can be reduced to the requirements based on "shared interests".

The idea is that when a person violates probabilism, they consist of several different "fragments". Each person-fragment *necessarily* obeys probabilism. The only reason divergence between person-fragments violates a rational norm is that there is normally a very high degree of "shared interests" and "capacity to co-ordinate" between the fragments.

- One worry about this is that fragmentation might apply to preferences as well as credences. In that case, assuming "interests" are understood in terms of preferences, the "shared interests" relation might not obtain between the fragments.

6. Ecological rationality

Kenny holds that according to one "conception" of rationality, the only rational requirements are those of probabilism along with (perhaps) logical omniscience and cross-temporal/interpersonal coherence. But in chapter 8, Kenny allows that there is also a

different “conception” of rationality on which it is much more demanding, so that (e.g.) probabilistically coherent conditionalizers with wacky counterinductive priors may count in this sense as irrational (violating “requirements”).

This less-permissive notion of rationality is what Kenny calls ‘ecological rationality’. It works in a highly externalistic way: ‘one doxastic state is more ecologically rational for a person than another if it leads that person, in their home environment, to perform actions that have a higher objective expected value than those actions that the other credence function would lead them to perform’.

- If we understand ‘objective expected value’ as an appeal to *objective chance*, then we can say that someone who has high credence in the truth about what the stock market will do next year is not very ecologically rational: they will *in fact* make a killing, but their actions don’t have a high *chance* of making them a killing and thus do not have a high objective expected value. But I gather Kenny is pretty sceptical about the concept “objective chance”; so how are we to take this?
- Anyway, it seems that the most ecologically rational doxastic states will involve having credence 1 in all propositions that in fact have chance 1, such as the true laws of physics.
 - If so, it seems that Kenny’s view is pretty revisionary with respect to our ordinary rationality judgments! We’d ordinarily think that someone who was certain of some particular theory of physics (without having done many experiments) was *less* rational than someone who divided their credence between many different theories. But on one of Kenny’s two conceptions (the permissive one), they are both equally rational, while on the other one (the ecological one), the one who’s certain is *more* rational so long as the credences are correct.

7. The perspective-relativity of ecological rationality

One important feature of Kenny’s notion of ecological rationality is that it is *perspective-relevant*. Claims of ecological rationality are only true or false relative to certain parameters: ‘To judge one of these dispositions as more or less ecologically rational than the other, we would need to say which range of worlds matter, and the relevant importance of getting at the truth or avoiding incorrectness’ (12). ‘There is no objectively correct environment from which we have to evaluate ecological rationality.’

- Can this help with the problem of people who are certain about physics? Could Kenny maybe say that they are highly ecologically rational *relative to the set of all physically possible worlds*, but irrational *relative to the set of all metaphysically possible worlds*, and philosophers who judge them “irrational” are just fixating on that set, or some other big set? But how do we understand ecological rationality relative to a set of worlds? One might say that the ecologically rational credence-forming methods

are those that lead to the truth in “most” worlds in the set—but this talk of “most” makes sense only relative to an antecedently given probability measure on the set of worlds!

- One could try avoiding this by relativizing not to a *set* of worlds but to a *probability function*. One’s methods are ecologically-rational-relative-to-Pr iff they have high Pr of leading to actions with high utility. Roughly speaking, it’ll then turn out that one is maximally ER-relative-to-Pr iff one has Pr as one’s ur-prior credence function.
- But, when people try to inquire into what credences are “most rational” given certain evidence, how is the relevant Pr-parameter to be filled in? They may already know that Pr_1 gives higher probability to H_1 and Pr_2 gives higher probability to H_2 . They *think* they are trying to answer a question ‘Which of Pr_1 and Pr_2 is more rational?’ which cannot be trivially resolved by pointing out that Pr_1 is the most ecologically-rational-relative-to- Pr_1 and Pr_2 is the most ecologically-rational-relative-to- Pr_2 . Are they just confused?